COMP3114 Information System Project / CSAI3124 Artificial Intelligence Application Project

System Specification

Online Shopping System

Overview

The Web (including the Mobile Web) has provided a unique opportunity for retail businesses. Customers may shop over a wide range of products and services without limitation on business hours and transportation. Vendors save huge expense in operating bricks-and-mortar shops and can reach a large number of potential customers, providing services for both large and niche markets. Online shopping malls like Taobao, Amazon and (to a lesser degree) eBay connect customers with vendors which offer a wide range of products. Big consumer brands host their own online store to sell tangible products (e.g. mobile phones), downloadable products (e.g. mobile apps) and virtual products (e.g. e-tickets). In addition, open-source e-commerce systems (e.g. Magento, osCommerce and OpenCart) provide low-cost solutions to small business to set up an online store.

This "Online Shopping System" is aimed at providing an online shopping experience to customers. The vendor displays their products in a manner that is easy for potential customers to select and make purchases.

Scope and workload structure

This system focuses on facilitating the online shopping experience. Most of the functional requirements are about the shopping experience visible to customers. This includes browsing and maintaining the product catalogue, and placing and processing purchase orders.

To limit the scope, this project omits payment processing and most of inventory management. The delivery process is assumed to be instantaneous. Right after a vendor ships a purchase a customer can receive the purchase.

While the preferred group size for this project is six, actual group sizes may vary due to factors such as add/drop periods or class sizes not being divisible by six. Accordingly, the assigned workload scales in proportion to the number of group members. The workload is divided into units of system requirements called "blocks." There are two categories of requirements: basic and further requirements. Basic requirements (blocks A-F) are mandatory, and projects that do not fully meet them will be considered incomplete. Further requirements (blocks S-Z), while not compulsory, must be completed by groups seeking a higher mark. The workload is summarized in Table 1.

Table 1. Summary of workload

Group size	Basic requirements	Further requirements
4	A, B	1 from W-Z, plus 1 more from S-Z
5	A, B, plus 1 from C-F	1 from W-Z, plus 2 more from S-Z
6	A, B, plus 2 from C-F	1 from W-Z, plus 3 more from S-Z

Basic requirements

All the basic functional requirements are organized into blocks as listed below. Details are described in a latter section of this document. All groups must complete basic requirement blocks A and B. A group of size above four has to complete more blocks. For example, a group of size 6 must complete blocks A and B, and any two blocks from C, D, E and F. In the report, students have to clearly state the basic requirement blocks they have completed.

- Block A: basic features to support minimal online shopping functions
- Block B: multiple photos per product and basic order processing
- Block C: more product attributes and advanced product search
- Block D: configurable products and preliminary inventory management
- Block E: split order and partial shipment
- Block F: multiple vendors (and shops)

Further requirements

On top of the basic requirements, students are encouraged to attempt *further requirements*. The further requirements in this specification consist of four advanced features (blocks S, T, U and V) and four non-functional requirements (blocks W, X, Y and Z) as listed below. In general, this specification provides less detailed and less concrete information on the further requirements. The students are expected to carry out more background study and system analysis before development, and to explain the detail in the final report. In addition, some of these further requirements involve cross-cutting issues that affect the design and implementation of the basic requirements.

- Block S: product recommendations
- Block T: user generated content
- Block U: wish list and promotional pricing strategy
- Block V: reports and analytics
- Block W: accessibility
- Block X: user experience (UX) design
- Block Y: search engine optimization
- Block Z: web application security

The number of further requirements that a project group should attempt is summarized in Table 1. A project group of any group size should elaborate on at least one non-functional requirement (blocks W to Z). In general, a larger group should attempt more of the further requirements.

AI application requirement

Project groups in CSAI3124 are required to investigate the application of artificial intelligence (AI) within the problem domain and to design and implement AI-powered functionalities in their system. Students should leverage AI to enhance both the basic and further requirements. Examples include recommending products by analysing user behaviours and interests, deploying chatbots to help users find products they are likely to want to buy, and using automated moderation to manage user reviews and comments for quality and relevance.

Detail of requirement blocks

Block A – Core functions

The e-commerce system has two main kinds of users. *Customers* do shopping in the store front, and *vendors* manage the product catalogue and process orders in the administration portal.

The store front provides four main functions to customers: product search, product detail information, shopping cart and view orders. The interface should be designed to be easy to use and attractive to customers.

A1. Customers may **register a new account**. They have to provide full name, email address, password and shipping address in registration.

A2. The product search and product detail information are accessible to customers before login. After login, customers can, in addition, manage their shopping carts and orders. Shopping cart content, purchase orders and other customer information are stored in the server side and are persisted across user sessions.

A3. Customers may browse numerous products in a list (or grid, if appropriate). This **product list** shows basic information of products, including product name, price and a thumbnail image.

A4. A customer can **search products** by product name (and optionally other product attributes). The search result is shown in a list.

A5. Customers are provided suitable controls for **navigating a long product list**. For example, desktop interfaces typically support pagination, while mobile interfaces usually employ infinite scroll or a 'load more' button.

A6. Customers may click/tap a product in the product list to view **product detail information**, which includes two more attributes in addition to the product name, price and a thumbnail image.

A7. Customers can **add** a product to their **shopping carts**. The default quantity to buy is 1.

A8. Customers can **list** the products in their **shopping carts**, showing the product name, price and quantity to buy. The total order amount (i.e. how much the customer has to pay in total) in a shopping cart is also shown. A customer can click/tap an item in the shopping cart to review the product detail.

A9. Customers can **change the quantity** to buy for a product in the **shopping carts**, if it makes sense for the kind of products sold in the e-commerce system.

A10. Customers can **remove** a product from their **shopping carts**.

A11. Customers can **check out** all items in the shopping cart. This action creates a purchase order and clears the content of the cart.

A12. Customers can **list** the **purchase orders** that they have placed. The following is shown for each purchase order: the P.O. number, the purchase date, the total order amount and the purchase order status. The purchase orders are displayed in reverse chronological order of purchase date. When a customer clicks/taps an entry in the list, they can see the detail of the purchase order.

A13. The **detail of a purchase order** to show includes the P.O. number, the purchase date, the shipping address, the total order amount and the purchase order status. For each product in the purchase order, the product name, the quantity, the unit price and the subtotal are displayed.

The administration portal is used by the administrator or a vendor to manage the product catalogue and purchase order processing. For basic version of this specification, no user authentication is required for the administration interface. The user interface should be designed to display lots of information clearly and to enable convenient administration operation.

A14. The vendor can **browse** the products in the **product catalogue**, and **search products** by product name (and optionally other product attributes).

A15. A system-generated unique **product ID** is also shown. The vendor can search products by entering a substring of the product ID.

A16. The vendor may **add a new product** to the catalogue. The vendor enters detail information of the product, which includes, but is not limited to, the product name, price and a thumbnail image.

A17. The vendor can edit product detail information.

A18. The vendor can **disable a product** to temporarily hide it from customers in the store front. The vendor can later enable the product.

A19. The vendor can **list the purchase orders** placed by customers. The purchase orders are sorted in descending order of purchase date (i.e. newest first). The list shows the P.O. numbers, purchase dates, customer names, total order amounts and purchase order status.

A20. The vendor can click/tap a **purchase order** to show its **detail** and line items.

Block B – Multiple photos and order processing

Block B includes support of more than one photo for each product and basic order processing.

B1. The e-commerce system supports **multiple photos** per product. The store front displays the photos in product detail, and the administration portal allows adding and removing photos from a product. (Optional: some products may use a short video in addition to photos.)

B2. The e-commerce system supports **basic order processing**. An order status is maintained for each purchase order to track the order processing workflow. You can define your own **order status workflow**, which contains at least four possible values for order status. The store front and the administration portal allow the customers or vendors to carry out actions that trigger order status change. Two examples are provided below.

- For tangible products: pending (a customer just places an order), shipped (the vendor ships a pending order), cancelled (a customer cancels a pending order before it is shipped) and hold (the vendor cannot ship the order for some reasons, and need to

revise the order). The vendor can ship a hold order later after resolving some issues. The customer can also cancel a hold order.
    - For virtual products (e.g. e-tickets): pending (a customer just places an order), ticket-issued (the vendor issues e-tickets for a pending order), complete (the e-tickets have been used, or have passed the refundable period), and refunded (the customer applies and receives refund within the refundable period).

B3. Customers can **filter** the list of **purchase orders** by order status.

B4. The system keeps record of the **date for some changes of order status**, and displays these dates together with other order detail information. Example includes shipment date, cancel date, ticket issue date and refund date.

Block C – Advanced product search and navigation

This block facilitates browsing a large product catalogue and allows narrowing down the search results with multiple criteria or advanced search.

C1. Product detail information includes **more attributes** than those in (A3). The attributes should be appropriate for the kind of products being sold in the e-commerce system. There is at least one attribute that support basic HTML formatting, including paragraphs and list.

C2. Product keyword **search** is done across **more than one** product attributes.

C3. Customers can **filter the search results** of (C2) in one of the following ways:
- category and price range, e.g. "gaming laptop" from $10000 to $15000
- one or more tags, e.g. #gaming #laptop #entry-level

C4. Under the detail information of a product, the store front shows brief description of **related products**. These related products are chosen with respect to product attributes. For example, the store front can list the books written by the same author, or other laptop computers similar in specification. Customers can click / tap to check the detail information of a related product.

C5. The product catalogue in the administration portal supports **editing** of the above **attributes** in product detail information.

Block D – Configurable products

This block includes support of configurable products and preliminary inventory management.

D1. In addition to simple products (which has no options), the vendor can **create configurable products** with at least 1 option. Each option has a predefined set of values (e.g. shirt size)

D2. Customers must choose a value for each available option when they **put a configurable product into shopping cart**. For example, a customer selects the colour and size of a shirt. A photo is shown to differentiate at least one option, e.g. a photo of a shirt of the selected colour regardless of the size.

D3. Customers can buy more than 1 items of a certain configuration of the product. In addition, they can buy **several configurations of the product**, e.g. 2 white T-shirt and 1 red T-shirt.

D4. In the administration portal, the product catalogue keeps separate **SKU** (stock keeping unit) for different configurations of a configurable product. A simple product has only one SKU. The vendor can set the **inventory status** of each SKU as in-stock or out-of-stock.

D5. A simple product or a certain configuration of a configurable product may be **out-of-stock**. In this case, customers cannot add the product to the shopping cart. (Optional: handle the case that a product in the shopping cart becomes out-of-stock just before checkout.)

Block E – Split orders and partial shipping

This block supports split orders and partial shipping. It requires adjustment in order processing in Block B and preliminary inventory management in Block D.

E1. An order may be **partially shipped**. For example, three out of five line items in an order may be shipped while the remaining two are temporarily out-of-stock. The e-commerce system keeps separate status for each line item in a purchase order.

E2. An order may be fulfilled in **more than one shipment**. The vendor can check shipments done for a purchase order, and which line items have not been shipped.

E3. Customers can check the shipments done for a purchase order, and which line items have not been shipped in an order.

E4. If some items are not yet shipped in an order, customers can **partially cancel** the order and apply refund.

Block F – Multiple vendors

This block supports multiple vendors, where each vendor can manage their own product catalogue and purchase orders. Each vendor manages a shop in the store front.

F1. A vendor can register a new account to **open a shop**. They have to provide the contact information of the person-in-charge and the shop information with photos.

F2. A vendor manages their own product catalogue in their shop, and processes the purchase orders and shipment in the administration portal.

F3. Customers can **visit the shop** and see the shop information, followed by some products in the shop. Customers can list the products in a shop and search products within the shop.

F4. Customers can find products from different shops in the e-commerce system, and put the products in their shopping cart. However, a purchase order can contain products from one shop only. Customers can select items in their shopping carts to check out, and the system will split the items into separate purchase order for each shop. (**split order by shop**)

F5. Items that are not checked out remain in the shopping carts.

Block S – Product recommendations

Simplifying product discovery for customers is a huge advantage. The recommended products can increase the chance that a customer will find relevant products, which can lead to increased sales.

Product recommendation seeks to predict and show the items that a user might like to purchase. Examples of recommendation strategies can be based on products that are commonly purchased together, have a similar product description, being best-sellers, frequently purchased, etc.

Block T – User generated content

User-generated content (UGC) is a powerful way to promote sales. Examples of UGC include customer feedback such as rating and/or writing a short review for a purchased product. Consumers trust content created by customers more than content created by brands. You should decide how to collect UGC and leverage it to promote your products.

Block U – Wish list and promotional pricing strategy

Customers can create a list of desired products to be purchased sometime in the future. Discounts and special offers can be planned around products in Wishlist to encourage purchase. Informing customers that items in their wish list have gone on sale is an excellent tactic to pull them back to complete the purchase. Moreover, the system can monitor search queries and search result browsing to anticipate potential sales.

Block V – Reports and analytics

Product sales can offer insights or business intelligence. Sales reports can help the vendor to glance at daily, weekly, monthly or annual sales numbers to understand the pace of growth. From weekly sales volume to top-performing products, everything that is sales critical is of great help to the vendor. Design suitable sales reports for online business and test your design with sample data.

Block W – Accessibility

Accessibility is important for individuals, businesses, society in order to provide equal access and equal opportunity to people with diverse abilities. Accessibility supports social inclusion for people with disabilities as well as others, such as older people, people with limited visibility abilities. When you design your system, you should study the challenges in streamlining user experience for people with diverse abilities, consult established guidelines (e.g. Web Content Accessibility Guidelines), and explore relevant assistive technologies (e.g. text-to-speech).

Another common requirement in making web applications accessible to wider audience is internationalization. Internationalization, sometimes referenced as i18n, is the process of designing and preparing your software products for use in different locales around the world. Examples of such process include text translation into different languages, formatting date and time according to specific locale, etc.

Businesses can no longer afford to limit content to their default language. Internationalization considerations enable users to interact with the software in their native mother tongue, which improves the user experience and overall diversity, inclusion, and accessibility of the software. In addition, with internationalization, new customers and markets become available.

Block X – User experience (UX) design

User Experience (UX) design refers to the process of creating a web/mobile application that is intuitive, user-friendly, and provides a positive experience for its users. It involves understanding the needs, goals, and expectations of the target users and designing the application in a way that meets those needs effectively.

Users of a web application or a mobile application have different expectation of the system interaction. Moreover, web and mobile applications possess distinct UX characteristics that you should take in consideration in system design. For example, desktop web applications can use a larger screen estate to display more complicated information, and support ergonomic UI to handle a large number of transactions. However, the browser window size may be different across user sessions. Responsive design techniques can be used to optimize the storefront's content and layouts to make it render well on all screen sizes and resolutions to improve overall user experience and satisfaction, especially in a variety of situations, across different devices.

Mobile applications, on the other hand, usually run on more limited screen estate. Besides having a mobile-friendly web application for customers to make a purchase from any device, consider having a mobile app for the shopping system. Mobile app, having the app icon/brand logo on the smart devices, helps to keep your brand on your customer's minds. In addition, with push notifications, you don't have to wait for customers to open up a browser to get your latest promotional messages. The mobile app should communicate with the server backend through a Web API and you have to design and implement both the frontend and the backend. You should also design the UX of the mobile application to meet users' expectation.

Notifications are powerful tools to inform users of relevant messages or introduce them to new features and updates. They help connect with users and promote engagement but can also result in user annoyance. Design and manage relevant notifications to get it right for better overall user experience.

Block Y – Search engine optimization

Search engine optimization (SEO) can attract traffic and is essential to the success of any internet business. Ranking on search engines can be optimized, from search-friendly URLs to meta structure, heading tags, and structured data, etc. Pages of product information should be indexable by search engines, and the product information should be organized in a structure that search engines can understand and extract. You should describe what SEO strategies you have implemented and explain how these affect the design and implementation of the system.

Block Z – Web application security

Web application security is the process of securing web applications from security vulnerabilities and attacks, ensuring that the application is free from vulnerabilities that could allow hackers to access sensitive data, steal information, or disrupt the application's functionality. An unsecured web application could result in losing sensitive data or downtime, leading to consequences of lost sales, broken customer trust, etc. You should discuss and implement a few measures to secure your Web applications to guard against some of the common web application security vulnerabilities.